

- (1) Write the Matlab or Python code to compute the FFT of two N -point real signals x and y using only one N -point FFT. (20 scores)

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$$[Fx, Fy] = \text{fftreall}(x, y)$$

The code should be handed out by NTUCool.

```
Input real sequence f1[n]:
[1, 2, 3, 4]
Input real sequence f2[n]:
[4, 8, 7, 3]
DFT Result F1:
[10.+0.j -2.+2.j -2.+0.j -2.-2.j]
DFT Result F2:
[22.-0.j -3.-5.j -0.+0.j -3.+5.j]
```

- (2) Suppose that $\text{length}(x[n]) = 1200$. What is the best way to implement the convolution of $x[n]$ and $y[n]$ if

- (a) $\text{length}(y[n]) = 300$, (b) $\text{length}(y[n]) = 30$,
(c) $\text{length}(y[n]) = 8$, and (d) $\text{length}(y[n]) = 2$?

Please show (i) the calculation method (direct, non-sectioned convolution, or sectioned convolution), (ii) the number of points of the FFT, (iii) and the number of real multiplications for the best implementation method. Also, consider the general case where $x[n]$ and $y[n]$ are complex sequences and the FFT of $y[n]$ can be computed in prior. (25 scores)

(a) $\text{length}(y[n]) = 300$

direct:

$$3 \times N \times M = 3 \times 1200 \times 300 = 1080000$$

✓ non-sectioned convolution: $P \geq N + M - 1 = 1499$ (P 取 1680)

$$2 \text{MUL}_{1680} + 3 \times 1680 = 2 \times 10420 + 3 \times 1680 = 25880 \text{ (real MULs)}$$

sectioned convolution: $L_0 = 599$ (python 找出), $P_0 = L_0 + M - 1 = 898$

① $P = 1008$ $S \times (2 \text{MUL}_{1008} + 3 \times 1008)$ ② $P = 840$ $S \times (2 \text{MUL}_{840} + 3 \times 840)$

$L = P - M + 1 = 2 \times (2 \times 5356 + 3 \times 1008)$ $L = P - M + 1 = 3 \times (2 \times 4580 + 3 \times 840)$
 $= 709$ $= 27472$ $= 541$ $= 35040$

$S = \lceil \frac{1200}{709} \rceil = 2$ $S = \lceil \frac{1200}{541} \rceil = 3$

③ $P = 720$ $S \times (2 \text{MUL}_{720} + 3 \times 720)$ ④ $P = 560$ $S \times (2 \text{MUL}_{560} + 3 \times 560)$

$L = P - M + 1 = 3 \times (2 \times 3620 + 3 \times 720)$ $L = P - M + 1 = 5 \times (2 \times 3100 + 3 \times 560)$
 $= 421$ $= 28200$ $= 261$ $= 39400$

$S = \lceil \frac{1200}{421} \rceil = 3$ $S = \lceil \frac{1200}{261} \rceil = 5$

(b) $\boxed{\text{length}(y[n]) = 30}$

direct:

$$3 \times N \times M = 3 \times 1200 \times 30 = 108000$$

non-sectioned convolution: $P \geq N + M - 1 = 1229$ (P 取 1260)

$$2 \text{MUL}_{1260} + 3 \times 1260 = 2 \times 7640 + 3 \times 1260 = 19060$$

✓ sectioned convolution: $L_0 = 174$ (python 找出), $P_0 = L_0 + M - 1 = 203$

① $P = 240$	$S \times (2 \text{MUL}_{240} + 3 \times 240)$	② $P = 204$	$S \times (2 \text{MUL}_{204} + 3 \times 204)$
$L = P - M + 1 = 211$	$= 6 \times (2 \times 940 + 3 \times 240)$	$L = P - M + 1 = 175$	$= 7 \times (2 \times 976 + 3 \times 204)$
$S = \lceil \frac{1200}{211} \rceil = 6$	$= 15600$	$S = \lceil \frac{1200}{175} \rceil = 7$	$= 17948$

③ $P = 168$	$S \times (2 \text{MUL}_{168} + 3 \times 168)$	④ $P = 144$	$S \times (2 \text{MUL}_{144} + 3 \times 144)$
$L = P - M + 1 = 139$	$= 9 \times (2 \times 580 + 3 \times 168)$	$L = P - M + 1 = 115$	$= 11 \times (2 \times 436 + 3 \times 144)$
$S = \lceil \frac{1200}{139} \rceil = 9$	$= 14976$	$S = \lceil \frac{1200}{115} \rceil = 11$	$= 14344$ (real MULs) ✖

⑤ $P = 120$	$S \times (2 \text{MUL}_{120} + 3 \times 120)$	⑥ $P = 96$	$S \times (2 \text{MUL}_{96} + 3 \times 96)$
$L = P - M + 1 = 91$	$= 14 \times (2 \times 380 + 3 \times 120)$	$L = P - M + 1 = 67$	$= 18 \times (2 \times 280 + 3 \times 96)$
$S = \lceil \frac{1200}{91} \rceil = 14$	$= 15680$	$S = \lceil \frac{1200}{67} \rceil = 18$	$= 15264$

(c) $\boxed{\text{length}(y[n]) = 8}$

direct:

$$3 \times N \times M = 3 \times 1200 \times 8 = 28800$$

non-sectioned convolution: $P \geq N + M - 1 = 1207$ (P 取 1260)

$$2 \text{MUL}_{1260} + 3 \times 1260 = 2 \times 7640 + 3 \times 1260 = 19060$$

✓ sectioned convolution: $L_0 = 30$ (python 找出), $P_0 = L_0 + M - 1 = 37$

① $P = 48$	$S \times (2 \text{MUL}_{48} + 3 \times 48)$	② $P = 36$	$S \times (2 \text{MUL}_{36} + 3 \times 36)$
$L = P - M + 1 = 41$	$= 30 \times (2 \times 92 + 3 \times 48)$	$L = P - M + 1 = 29$	$= 42 \times (2 \times 64 + 3 \times 36)$
$S = \lceil \frac{1200}{41} \rceil = 30$	$= 9840$	$S = \lceil \frac{1200}{29} \rceil = 42$	$= 9912$

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$$\begin{array}{ll}
 \textcircled{3} \quad P = 24 & S \times (2 \text{MUL}_{24} + 3 \times 24) \\
 L = P - M + 1 = 17 & = 71 \times (2 \times 28 + 3 \times 24) \\
 S = \lceil \frac{1200}{17} \rceil = 71 & = 9088 \text{ (real MULs)} *
 \end{array}
 \quad
 \begin{array}{ll}
 \textcircled{4} \quad P = 12 & S \times (2 \text{MUL}_{12} + 3 \times 12) \\
 L = P - M + 1 = 5 & = 240 \times (2 \times 8 + 3 \times 12) \\
 S = \lceil \frac{1200}{5} \rceil = 240 & = 12480
 \end{array}$$

(b) $\boxed{\text{length}(y[n]) = 2}$

direct:

$$3 \times N \times M = 3 \times 1200 \times 2 = 7200$$

non-sectioned convolution: $P \geq N + M - 1 = 1201$ (P 取 1260)

$$2 \text{MUL}_{1260} + 3 \times 1260 = 2 \times 7640 + 3 \times 1260 = 19060$$

✓ sectioned convolution: $L_0 = 2$ (python 找出), $P_0 = L_0 + M - 1 = 3$

$$\begin{array}{ll}
 \textcircled{1} \quad P = 4 & S \times (2 \text{MUL}_4 + 3 \times 4) \\
 L = P - M + 1 = 3 & = 400 \times (2 \times 0 + 3 \times 4) \\
 S = \lceil \frac{1200}{3} \rceil = 400 & = 4800 \text{ (real MULs)} *
 \end{array}
 \quad
 \begin{array}{ll}
 \textcircled{2} \quad P = 3 & S \times (2 \text{MUL}_3 + 3 \times 3) \\
 L = P - M + 1 = 2 & = 600 \times (2 \times 2 + 3 \times 3) \\
 S = \lceil \frac{1200}{2} \rceil = 600 & = 7800
 \end{array}$$

$$\begin{array}{ll}
 \textcircled{3} \quad P = 2 & S \times (2 \text{MUL}_2 + 3 \times 2) \\
 L = P - M + 1 = 1 & = 1200 \times (2 \times 0 + 3 \times 2) \\
 S = \lceil \frac{1200}{1} \rceil = 1200 & = 7200
 \end{array}$$

(3) (a) What are the number of entries equal to 1 and -1 for the 2^k -point Walsh transform? (b) What are the number of entries equal to 1, 0, and -1 for the 2^k -point Haar transform? (c) What is the most important application of the Walsh transform nowadays? (d) What is the most important advantage of the Haar transform nowadays? (20 scores)

(a) 對於 2^k -point 的 Walsh transform,

等於 1 的值有 $2^{k-1}(2^k+1)$ 個

等於 -1 的值有 $2^{2^k} - 2^{k-1}(2^k+1)$ 個 *

(b) 對於 2^k -point 的 Haar transform,

等於 1 的值有 $k \times 2^{k-1} + 2^k = k2^{k-1} + 2^k$ 個

等於 -1 的值有 $k \times 2^{k-1} = k2^{k-1}$ 個 *

(c) CDMA $\leadsto$ using the basis (rows) of the Walsh transform to perform modulation

(d) Analysis of the local high frequency component.

- (4) (a) What is the results of CDMA if there are three data $[1\ 0\ 1]$, $[1\ 1\ 0]$, $[0\ 1\ 1]$ and these three data are modulated by the 1st, 4th, and 10th rows of the 16-point Walsh transform? (The beginning row is the 1st row). (10 scores)
- (b) In (a), if the 7th and the 19th entries of the CDMA results are missed, can we recover the original data? Why? (5 scores)

(a) $[1\ 0\ 1] \rightarrow [1\ -1\ 1]$ $[1\ 1\ 0] \rightarrow [1\ 1\ -1]$ $[0\ 1\ 1] \rightarrow [-1\ 1\ 1]$

$$V_{16} = \begin{bmatrix} W_8 & W_8 \\ W_8 & -W_8 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \end{bmatrix}$$

Zero crossings

0
3
4
7
8
11
12
15

1
2
5
6
9
10
13
14

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix} + \begin{bmatrix} -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 3 & 3 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

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(b)

channel 1 $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
 $\Rightarrow \frac{15}{16} > 0 \Rightarrow 1$ $\Rightarrow \frac{-15}{16} < 0 \Rightarrow -1$ $\Rightarrow \frac{16}{16} = 1$ original data is $[1\ 0\ 1]$

channel 2 $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
 $\Rightarrow \frac{17}{16} > 0 \Rightarrow 1$ $\Rightarrow \frac{17}{16} > 0 \Rightarrow 1$ $\Rightarrow \frac{-16}{16} = -1$ original data is $[1\ 1\ 0]$

channel 3 $\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}$
 $\Rightarrow \frac{-15}{16} < 0 \Rightarrow -1$ $\Rightarrow \frac{15}{16} > 0 \Rightarrow 1$ $\Rightarrow \frac{16}{16} = 1$ original data is $[0\ 1\ 1]$ *

(5) Ramanujan's Sum in NTT

Given $M = 11$, $\alpha = 8+6i$, and $N = 12$. Please determine the complex number theoretic transform (CNT) of $\mathbf{x}$ if

$$\mathbf{x} = [0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1]$$

Hint: $\text{fft}(\mathbf{x})$ is as follows, which is Ramanujan's Sum

$$\text{fft}(\mathbf{x}) = [4 \ 0 \ 2 \ 0 \ -2 \ 0 \ -4 \ 0 \ -2 \ 0 \ 2 \ 0] \quad (8 \text{ scores})$$

$\therefore \text{gcd-delta} \xrightarrow{\text{NTT}} \text{Ramanujan's Sum}$, $\mathbf{x} = [0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1]$ is gcd-delta

$\therefore$ The complex number theoretic transform of $\mathbf{x}$ is $[4 \ 0 \ 2 \ 0 \ 2 \ 0 \ 4 \ 0 \ 2 \ 0 \ 2 \ 0]$ *

(6) (a) Please determine

$$3^{2049} \bmod 103 \quad (\text{Hint: 費馬小定理})$$

(b) Suppose that $x \bmod 43 = 2$ and $x \bmod 67 = 13$

Please Determine

$$x \bmod 2881. \quad (\text{Hint: Chinese Remainder Theorem})$$

(c) $n! = n(n-1)(n-2) \dots 1$. Please determine $39! \bmod 43$

(Hint: Wilson's Theorem)

(12 scores)

$$(a) \quad 3^{102} \equiv 1 \pmod{103}$$

$$3^{2049} = 3^{102 \cdot 20 + 9} = (3^{102})^{20} \cdot 3^9 = 1^{20} \cdot 3^9 \equiv 10 \pmod{103}, \text{ 所以 } 3^{2049} \bmod 103 = 10 *$$

$$(b) \quad m = 43 \times 67 = 2881, M_1 = 2881/43 = 67, M_2 = 2881/67 = 43$$

$$x = 2 \times 67 \times 9 + 13 \times 43 \times 53 = 30833, \text{ 所以 } 30833 \bmod 2881 = 2023 *$$

$$(c) \quad 42! \equiv -1 \pmod{43}, 42 \times 41 \times 40 \times 39! \bmod 43,$$

$$6 \times 39! \equiv 1 \pmod{43} \leadsto 6 \text{ 的 inverse}, \text{ 所以 } 39! \bmod 43 = 36 *$$

(Extra): Answer the questions according to your student ID number.

(ended with (1, 6), (2, 7), (3, 8), (4, 9))

$$Q: 5 \oplus 3 = ?$$

$$\begin{array}{r} 5 \ 101 \\ \oplus 3 \ 011 \\ \hline 6 \ 110 \end{array} \Rightarrow 5 \oplus 3 = 6 *$$